# Simulating the spread of the Black Death

**FYTN03**

*Author:*

Fredrik Bergelv
fredrik.bergelv@live.se

Lund University, Department of Physics

under the direction of Victor Olariu




LUND UNIVERSITY

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# 1 Introduction

In 1350, the bubonic plague, known as the Black Death, hit Scania [3]. Scania was an important region in the herring trade for the Hansiatic league, which meant that Scania had well established trade routes with cities such as Lübeck. It is therefor a possibility that the Black Death arrived to Scania from Lübeck. This article goes through different numeric models which will be used to simulate the spread of the Black Death between Lübeck and Scania, and view the theoretical effects which vaccination or quarantine would have done for the two regions.

# 2 Theory

There exist several models for describing the spread of diseases. One such model is the SIS model which describes the relation between infected $I$ and susceptible $S$ as

$$\begin{cases} \frac{\mathrm{d}I}{\mathrm{d}t} = \beta\frac{SI}{N} - \alpha I \\ \frac{\mathrm{d}S}{\mathrm{d}t} = -\beta\frac{SI}{N} + \alpha I. \end{cases} \tag{1}$$

Above $\beta$ denotes the infection rate, $\alpha$ denotes the rate of recovery and $N$ is the size of the population. The rates are conventionally chosen to be per day. After a long time however the model should reach an equilibrium state where $\frac{\mathrm{d}I}{\mathrm{d}t} = 0$. This gives the following relation, where we know that $S + I = N$.

$$0 = \frac{\mathrm{d}I}{\mathrm{d}t} = \beta\frac{SI}{N} - \alpha I \iff \beta\frac{SI}{N} = \alpha I \iff \frac{S}{N} = \frac{\alpha}{\beta} \iff \frac{I}{N} = 1 - \frac{\alpha}{\beta}. \tag{2}$$

The SIR model takes into account that people who have been infected develop a immunity against the disease and become recovered after some time, denoted by $R$. This models also ads a vaccination factor $\gamma$ which represents the factor of susceptible people who become immune by vaccination (or quarantine) each day. The differential equations then becomes

$$\begin{cases} \frac{\mathrm{d}I}{\mathrm{d}t} = \beta\frac{SI}{N} - \alpha S \\ \frac{\mathrm{d}S}{\mathrm{d}t} = -\beta\frac{SI}{N} - \gamma S \\ \frac{\mathrm{d}R}{\mathrm{d}t} = \gamma S + \alpha I. \end{cases} \tag{3}$$

The SIR model above only takes into account one population $N$, however in the description in section 1 we have two populations. One can therefore modify the equations above to include travel between two or more populations. The differential equations for city $n$ then become

$$\begin{cases} \frac{\mathrm{d}I_n}{\mathrm{d}t} = \beta\frac{S_nI_n}{N} - \alpha S_n + \sum_{m\neq n}\left[\omega_{n\leftarrow m}I_m - \omega_{m\leftarrow n}I_n\right] \\ \frac{\mathrm{d}S_n}{\mathrm{d}t} = -\beta\frac{S_nI_n}{N} - \gamma S_n + \sum_{m\neq n}\left[\omega_{n\leftarrow m}S_m - \omega_{m\leftarrow n}S_n\right] \\ \frac{\mathrm{d}R_n}{\mathrm{d}t} = \gamma S_n + \alpha I_n + \sum_{m\neq n}\left[\omega_{n\leftarrow m}R_m - \omega_{m\leftarrow n}R_n\right], \end{cases} \tag{4}$$

where $\omega_{n\leftarrow m}$ is the factor of people going from population $m$ to population $n$ and vice versa.

The SIRD model includes death of a certain number of infected by the mortality rate $\mu$. This creates a new category of dead people denoted as $D$. This changes the differential equation into

$$\begin{cases} \frac{\mathrm{d}I_n}{\mathrm{d}t} = \beta\frac{S_nI_n}{N} - \alpha S_n - \mu I_n + \sum_{m\neq n}\left[\omega_{n\leftarrow m}I_m - \omega_{m\leftarrow n}I_n\right] \\ \frac{\mathrm{d}S_n}{\mathrm{d}t} = -\beta\frac{S_nI_n}{N} - \gamma S_n + \sum_{m\neq n}\left[\omega_{n\leftarrow m}S_m - \omega_{m\leftarrow n}S_n\right] \\ \frac{\mathrm{d}R_n}{\mathrm{d}t} = \gamma S_n + \alpha I_n + \sum_{m\neq n}\left[\omega_{n\leftarrow m}R_m - \omega_{m\leftarrow n}R_n\right] \\ \frac{\mathrm{d}D}{\mathrm{d}t} = \mu I_n. \end{cases} \tag{5}$$

To solve the models above we need the constants $\alpha$, $\beta$, $\mu$. If we assume that the Black Death spread pneumatically then the infection rate, per incubation time, is 2.8 to 3.5, where the incubation time of the plague was between 1 and 7 days [4][6]. Using the mean of both of these values we obtain a mean rate of infection to be $\beta = 0.79$. The mortality rate of the Black death was 30% to 75% [5]. This results in a recovery rate of $\alpha = 0.12$ and a mortality rate of $\mu = 0.13$. The population size of Lübeck was estimated to be 38 000 and the population size of Scania was estimated to be 150 000[1][2].

To estimate the truncation error of a differential equation solver one can use the formula

$$E_n^h = \frac{y_n^h - y_n^{2h}}{2^m - 1}, \tag{6}$$

where $h$ is the step size, $n$ is each step, $y$ is the value obtained and $m$ is related to the order of accuracy of the numerical method. For the Runga-Kutte method of the forth $m = 4$.

# 3 Method

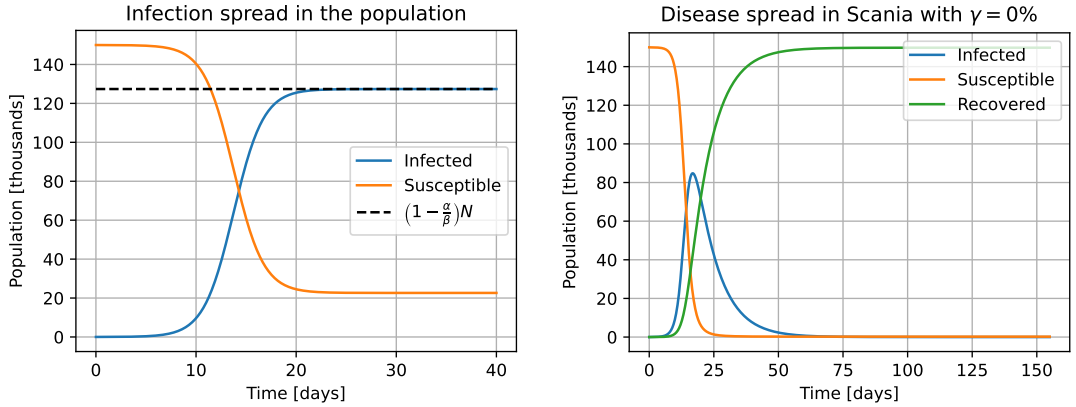
To solve the given differential equations presented, the Runge-Kutta method of the fourth and fifth order with a step size of $h = 10000$ was implemented. This method was chosen due to it's high order resulting in smaller error. This was done through a python code which can be seen in Appendix A. The python code used the solver `solve-ivp` from `scipy.integrate`.

The initial number of infected people in Lübeck was chosen to be 13, which was a normal size of crew of a merchant ship at the time. The travel factor $\omega$ was estimated by assuming that for each region two merchant ships of 13 people left each region and arrived in the other each day. This made the travelling factors to be $\omega_{\text{Scania}\leftarrow\text{Lübeck}} = 0.000\,68$ and $\omega_{\text{Lübeck}\leftarrow\text{Scania}} = 0.000\,17$. Vaccination was not yet invented during the 14th century. But a theoretical value of $\gamma = 0.02$ was chosen to study the affects of what a vaccine during this time could have done to the outcome of the black death.

The vaccination period was delayed for the SIRD model. This meant that the populations only started vaccinating people after a total of 100 people had been infected. Thus, people would not start to get vaccinated before it became well known that the disease had been detected in the population.

# 4 Result

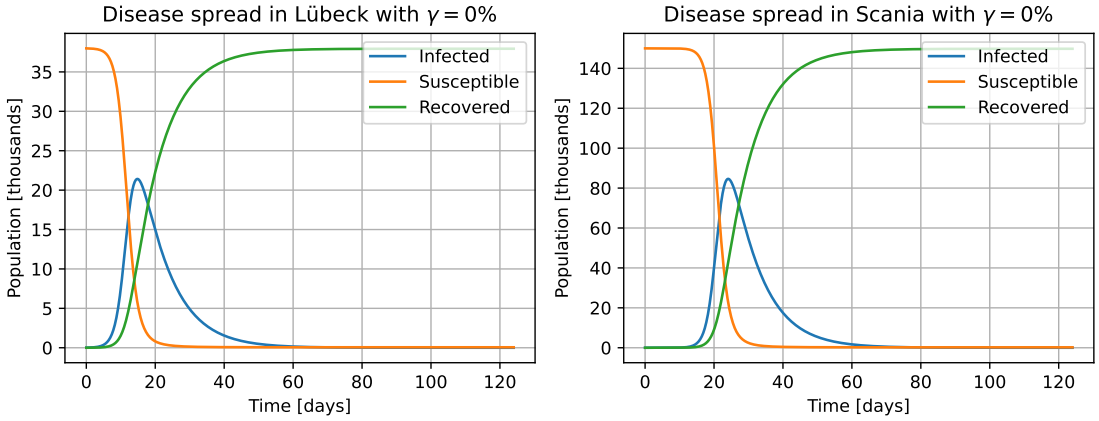
## 4.1 The SIS and SIR model



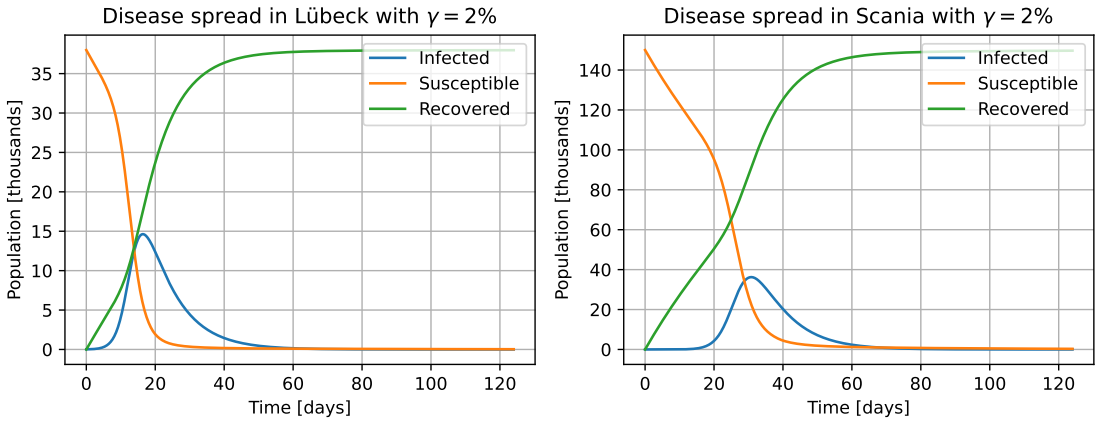

(a) This figure showsthe SIS modeland the theoretical fraction (Equation 2).

(b) This figure shows the SIR model without any vaccination factor $\gamma$ nor any travel.

Figure 1: Above one can se the SIS and the SIR model with $\beta = 0.79$ and $\alpha = 0.12$.




(a) This figure shows the outcome without any vaccination factor $\gamma$.




(b) This figure shows the outcome with a vaccination factor $\gamma = 0.02$.

Figure 2: Both figures show the STIR model with $\beta = 0.79$ and $\alpha = 0.12$.

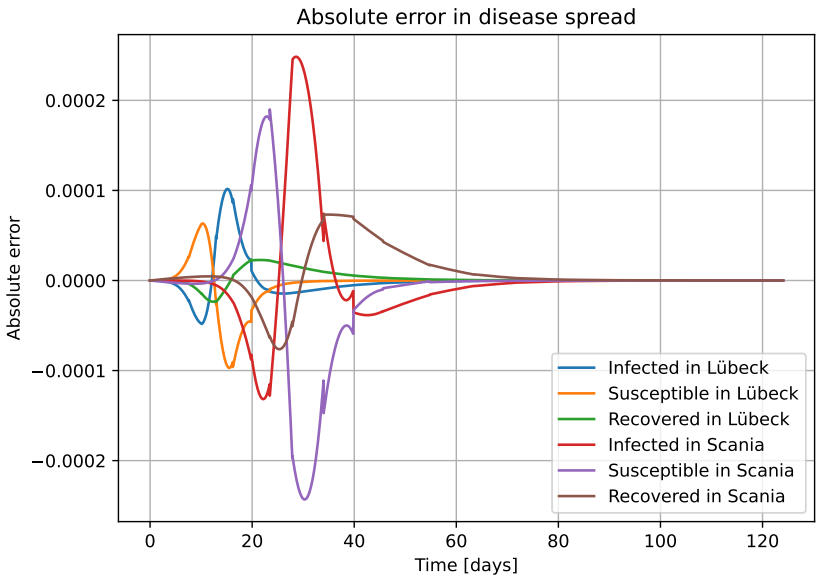
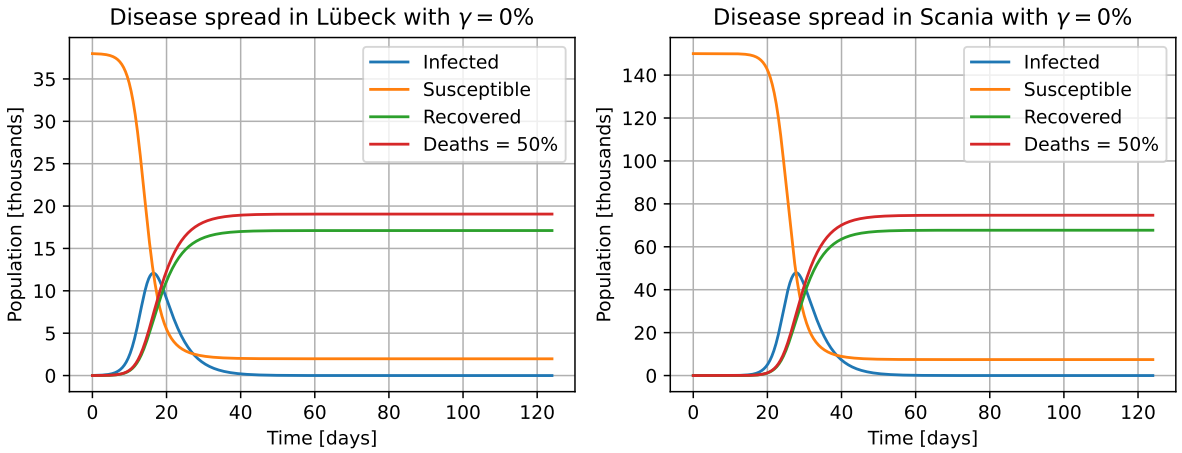


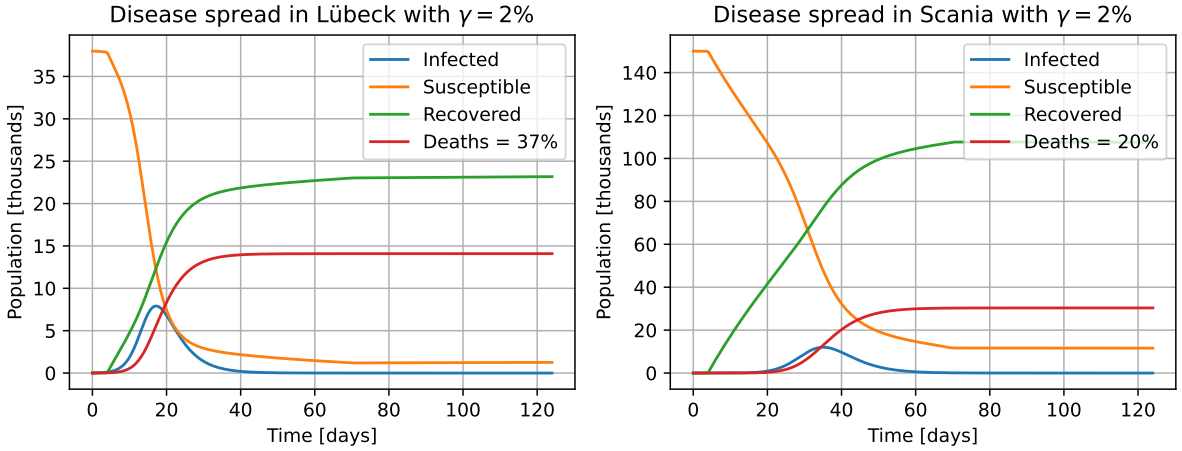

Figure 3: This figure shows the the absolute error for the SIR model shown in Figure 4b.

The SIS model showed a equilibrium fraction of 85% being infected. In the SIR model in Figure 1b immunity after being infected was introduced, which resulted in almost all being immune after about 50 days. With travel in Figure 2 the maximal fraction of infected without the introduction of any vaccination factor was 39% in both regions with a immunity in the end of 99%. The result differed between the two regions when a vaccination factor was introduced, then Lübeck had a peak infection fraction of 39% and Scania 24%, with a total immunity in the end of 99%.

## 4.2 The SIRD model




(a) This figure shows the outcome without any vaccination factor $\gamma$.




(b) This figure shows the outcome with a vaccination factor $\gamma = 0.02$ after 100 infected had been reached.

Figure 4: Both figures show the SIRD model with $\beta = 0.79$, $\alpha = 0.12$ and $\mu = 0.13$.

Without any vaccination the SIR model in Figure 4a showed an equilibrium after 40 days. At equilibrium 50% of both populations had died, and of the remaining population, 90% had recovered. A vaccination factor of $\gamma = 0.02$ after 100 total infected in Figure 4b resulted in a 26% decrease in deaths in Lübeck and a 60% decrease in deaths in Scania. Since the population of Scania is much larger then that of Lübeck, the vaccination led to 52% fewer deaths with a total of 24%. The fraction of infected in Lübeck and Scania was at its highest 39% and 40% without the vaccination, in comparison to 24% and 9% with the vaccination. If 2% of the non infected population went into quarantine per day the total death rate would thus drop by 52%.

# 5 Discussion

## 5.1 The SIS and SIR model

In the SIS model in Figure 1a the equilibrium number of infected people matches perfectly with the fraction which was predicted in Equation 2. In Figure 2 it is noticeable that travel between the two cities result in very similar result to that without any travel, with the only difference that the disease strikes Scania a bit later. Thus when a vaccination factor was added the peak infection fraction in Scania was lower then in Lübeck. However, it was lower in both regions compared to no vaccination factor. Both regions would thus function much better when fewer people would be sick at the same time. Another interesting result is the very high immunity after the infection. This means that if the same disease would hit the regions again, almost all would be immune.

The numeric errors in these models are really low, which can bee seen in Figure 3, where the absolute error is in the order of $10^{-4}$. However, it is still interesting that the error that is shown is in the middle of the infection period, but goes to zero afterwards. This is probably because the numeric solver is more uncertain what happens in the middle of the simulation but more certain of the en result, just like how the equilibrium could be solved quit easily for the SIS model in Equation 2 but the infection period was much harder to solve. Another interesting result in Figure 3 is that all the error functions are almost as much over the x-axis as they are under it. This means that when the solver overestimates something it compensates by underestimating it afterwards, which probably leads to a more accurate result.

However, the SIS and SIR are poor models when dealing with diseases with a high mortality rate. Both SIS and SIR assume that the number of deaths are small enough so they do not drastically affect the statistics, which is not the case with the Black Death. Also, the vaccination in the model starts before the big outbreak of the disease, which would not be the case in real life. Even if we assume that the vaccine is already available at the time of the first infected, people would not start to get vaccinated before it became well known that the disease had been detected in the population. This would offset the start of the vaccination, which would lead to a more rapid spread of the disease. In both cases the SIRD model is a better alternative.

One interesting observation is that immunity in the SIR model is much higher then in the SIRD model. The high immunity is a consequence of the high infection rate, but it is higher in the SIR model because all infected become immune afterwards. Since a large number of people get infected, it follows that almost none stay susceptible. In the SIRD model however a less people survive resulting in fewer immune.

## 5.2 The SIRD model

The SIRD model in Figure 4 showed half of the population being wiped out, which coincides with the historical data [3]. This model showed, like the SIR, high immunity afterwards. This is a interesting result, after a deadly pandemic the strong majority of the survivals will be immune to the disease. This means, that if the disease was to come back some years later the new population would handle the disease much better, since 90% would be immune.

In this model we also saw that the introduction of the vaccine lowered the peak fraction of infected people. This shows that the vaccination does not only save lives but also creates a more well functioning society where a much smaller percentage are sick. Even though they did not have vaccination during the 14th century, one could achieve the same result by going into quarantine.

However, the SIRD model in itself is not perfect. It does not take into account that variables may change. The vaccination rate (or quarantine rate) should vary depending on the number of infected. The percentage of people who go into quarantine or get vaccinated each day should increase when the number of infected increases due to the fear of getting infected yourself. The infection rate should also be lowered with the increase of the number of infected people, due to the fact that behavioural patterns of the population change when a large fraction are sick. In Figure 4a, a infection fraction of 40% would imply that the society shuts down, leading to less interactions and a lowered infection rate. These varying variables would further imply a lowering in the total number of deaths from the result obtained.

# 6 Conclusion

To summarise, the SIRD is a superior model when simulating the Black Death compared to the SIS or SIR models due to the fact that the Black Death had a high mortality rate. The SIRD showed a 50% total death rate with a remaining population of which 90% would be immune to the disease afterwards. The peak infection fraction of the population was 40%, which would lead to an collapse of society. However, by sending 2% of the population into quarantine each day (after a total of 100 infected had been reached) the total death rate was lowered to a total of 24% with a peak infection fraction of only 12%.

# References

[1] Hansestadt Lübeck. Statistische nachrichten nr. 41, April 2021. Statistical report.

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[3] Samuel K. Cohn JR. Epidemiology of the black death and successive waves of plague. *Medical History Supplement*, pages 74–100, 2008.

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[5] John Martin Rare Book Room. The black death: The plague, 1331-1770. Online exhibit, 2017. Published by Hardin Library for the Health Sciences, Iowa City, IA.

[6] World Health Organization. Plague, July 2022. Accessed: 2024-09-12.

# A The python code

```
"""
Created on Mon Sep 10 2024
@author: Fredrik Bergelv
"""
from scipy.integrate import solve_ivp
import numpy as np
import matplotlib.pyplot as plt

b = np.mean([2.8,3.5])/np.mean([1,7])   # Rate of infection
m = np.mean([30,75])/100/np.mean([1,7]) # Rate of mortality
g = 0.0#1*2                             # Rate of vaccination
a = 1/np.mean([1,7])-m                  # Rate of recovery

N_L = 38000
N_S = 150000

N_ship = 13
Number_ships_per_day = 2
w_SL = Number_ships_per_day*N_ship/N_S
w_LS = Number_ships_per_day*N_ship/N_L

I0_L, S0_L, R0_L, M0_L = N_ship, N_L-N_ship, 0, 0
I0_S, S0_S, R0_S, M0_S = 0, N_S, 0, 0

timescale = t0, t1 = 0, 31*4
h = 10000


def SIR(t, f):
    N=300000
    I, S, R = f
    dIdt = b*S*I/N-a*I
    dSdt = -b*S*I/N-g*S
    dRdt = g*S+a*I
    return dIdt, dSdt, dRdt

def SIR_travel(t, f):
    I_L, S_L, R_L, I_S, S_S, R_S = f

    g_L, g_S = g, g

    dI_L =  b*S_L*I_L/N_L - a*I_L + (w_SL*I_S-w_LS*I_L)
    dS_L = -b*S_L*I_L/N_L - g_L*S_L + (w_SL*S_S-w_LS*S_L)
    dR_L =  g_L*S_L + a*I_L + (w_SL*R_S-w_LS*R_L)

    dI_S =  b*S_S*I_S/N_S - a*I_S + (w_LS*I_L-w_SL*I_S)
    dS_S = -b*S_S*I_S/N_S - g_S*S_S + (w_LS*S_L-w_SL*S_S)
    dR_S =  g_S*S_S + a*I_S + (w_LS*R_L-w_SL*R_S)

    return dI_L, dS_L, dR_L, dI_S, dS_S, dR_S
    I_L, S_L, R_L, I_S, S_S, R_S = f
    g_L = g
    g_S = g

    dI_L =  b*S_L*I_L/N_L - a*I_L + (w_SL*I_S-w_LS*I_L)
```

```
    dS_L = -b*S_L*I_L/N_L - g_L*S_L + (w_SL*S_S-w_LS*S_L)
    dR_L =  g_L*S_L + a*I_L + (w_SL*R_S-w_LS*R_L)

    dI_S =  b*S_S*I_S/N_S - a*I_S + (w_LS*I_L-w_SL*I_S)
    dS_S = -b*S_S*I_S/N_S - g_S*S_S + (w_LS*S_L-w_SL*S_S)
    dR_S =  g_S*S_S + a*I_S + (w_LS*R_L-w_SL*R_S)

    return dI_L, dS_L, dR_L, dI_S, dS_S, dR_S

def SIRD_travel(t, f):
    I_L, S_L, R_L, M_L, I_S, S_S, R_S, M_S = f

    if I_L > 100 or I_S > 100:
        g_L = g
        g_S = g
    else:
        g_L = 0
        g_S = 0

    dI_L =  b*S_L*I_L/N_L - a*I_L - m*I_L + (w_SL*I_S-w_LS*I_L)
    dS_L = -b*S_L*I_L/N_L - g_L*S_L + (w_SL*S_S-w_LS*S_L)
    dR_L =  g_L*S_L + a*I_L + (w_SL*R_S-w_LS*R_L)
    dM_L =  m*I_L

    dI_S =  b*S_S*I_S/N_S - a*I_S - m*I_S + (w_LS*I_L-w_SL*I_S)
    dS_S = -b*S_S*I_S/N_S - g_S*S_S + (w_LS*S_L-w_SL*S_S)
    dR_S =  g_S*S_S + a*I_S + (w_LS*R_L-w_SL*R_S)
    dM_S =  m*I_S

    return dI_L, dS_L, dR_L, dM_L, dI_S, dS_S, dR_S, dM_S

#%% SIR
InitialCond1 = I0_L, S0_L, R0_L, I0_S, S0_S, R0_S

sol1 = solve_ivp(SIR_travel, [t0, t1], InitialCond1,
                 method='RK45', t_eval=np.linspace(t0, t1, h))

for i in range(0,6):
    sol1.y[i]=sol1.y[i]/1000
I_L, S_L, R_L = sol1.y[0], sol1.y[1], sol1.y[2]
I_S, S_S, R_S = sol1.y[3], sol1.y[4], sol1.y[5]
t = sol1.t

plt.figure(figsize=(15*0.6, 6*0.6))

plt.subplot(1, 2, 1)
plt.plot(t, I_L, label='Infected')
plt.plot(t, S_L, label='Susceptible')
plt.plot(t, R_L, label='Recovered')
plt.xlabel('Time [days]')
plt.ylabel('Population [thousands]')
plt.title(f'Disease spread in Lübeck with $\gamma={int(g*100)}\%$')
plt.grid(True)
plt.legend(loc='upper right')

plt.subplot(1, 2, 2)
plt.plot(t, I_S, label='Infected')
plt.plot(t, S_S, label='Susceptible')
```

```
plt.plot(t, R_S, label='Recovered')
plt.xlabel('Time [days]')
plt.ylabel('Population [thousands]')
plt.title(f'Disease spread in Scania with $\gamma={int(g*100)}\%$')
plt.legend(loc='upper right')
plt.grid(True)

plt.tight_layout()

def SIR_info(sol1):
    for i in range(0,5):
        sol1.y[i]=sol1.y[i]/1000
        I_L, S_L, R_L = sol1.y[0], sol1.y[1], sol1.y[2]
        I_S, S_S, R_S = sol1.y[3], sol1.y[4], sol1.y[5]

    for i in range (0,h):
        if I_L[i]==np.max(I_L):
            I_L_max = int(np.round(100*I_L[i]/(I_L[i]+S_L[i]+R_L[i])))
        if I_S[i]==np.max(I_S):
            I_S_max = int(np.round(100*np.max(I_S)/(I_S[i]+S_S[i]+R_S[i])))

        Imune = int(100*R_S[-1]/(S_S[-1]+R_S[-1]))

SIR_info(sol1)

sol_h = solve_ivp(SIR_travel, [t0, t1], InitialCond1,
                  method='RK45', t_eval=np.linspace(t0, t1, h))
sol_2h = solve_ivp(SIR_travel, [t0, t1], InitialCond1,
                  method='RK45', t_eval=np.linspace(t0, t1, h*2))

Titles = ('Infected in Lübeck', 'Susceptible in Lübeck',
          'Recovered in Lübeck','Infected in Scania',
          'Susceptible in Scania','Recovered in Scania')

m_err = 4
plt.figure(figsize=(7*0.8/0.5*0.6, 5*0.8/0.5*0.6))
for i in range(0,6):
    Err = (sol_h.y[i]-np.interp(sol_h.t,sol_2h.t,sol_2h.y[i]))/(2**m_err-1)
    plt.plot(sol_h.t, Err, label=Titles[i])
plt.xlabel('Time [days]')
plt.ylabel('Absolute error')
plt.title('Absolute error in disease spread')
plt.legend()
plt.grid(True)
plt.legend(loc='lower right')

plt.tight_layout()

#%% SIRD
InitialCond2 = I0_L, S0_L, R0_L, M0_L, I0_S, S0_S, R0_S, M0_S

sol2 = solve_ivp(SIRD_travel, [t0, t1], InitialCond2,
                  method='RK45', t_eval=np.linspace(t0, t1, h))

for i in range(0,8):
    sol2.y[i]=sol2.y[i]/1000
I_L, S_L, R_L, M_L = sol2.y[0], sol2.y[1], sol2.y[2], sol2.y[3]
I_S, S_S, R_S, M_S = sol2.y[4], sol2.y[5], sol2.y[6], sol2.y[7]
```

```
t = sol2.t

plt.figure(figsize=(15*0.6, 6*0.6))

plt.subplot(1, 2, 1)
plt.plot(t, I_L, label='Infected')
plt.plot(t, S_L, label='Susceptible')
plt.plot(t, R_L, label='Recovered')
plt.plot(t, M_L, label=f'Deaths = {round(100*np.max(M_L*1e3)/N_L)}%')
plt.xlabel('Time [days]')
plt.ylabel('Population [thousands]')
plt.title(f'Disease spread in Lübeck with $\gamma={int(g*100)}\%$')
plt.grid(True)
plt.legend(loc='upper right')

plt.subplot(1, 2, 2)
plt.plot(t, I_S, label='Infected')
plt.plot(t, S_S, label='Susceptible')
plt.plot(t, R_S, label='Recovered')
plt.plot(t, M_S, label=f'Deaths = {round(100*np.max(M_S*1e3)/N_S)}%')
plt.xlabel('Time [days]')
plt.ylabel('Population [thousands]')
plt.title(f'Disease spread in Scania with $\gamma={int(g*100)}\%$')
plt.legend(loc='upper right')
plt.grid(True)

plt.tight_layout()

def SIRD_info(sol2):
    for i in range(0,8):
        sol2.y[i]=sol2.y[i]/1000
    I_L, S_L, R_L, M_L = sol2.y[0], sol2.y[1], sol2.y[2], sol2.y[3]
    I_S, S_S, R_S, M_S = sol2.y[4], sol2.y[5], sol2.y[6], sol2.y[7]

    for i in range (0,h):
        if I_L[i]==np.max(I_L):
            I_L_max = int(np.round(100*I_L[i]/(I_L[i]+S_L[i]+R_L[i])))
        if I_S[i]==np.max(I_S):
            I_S_max = int(np.round(100*np.max(I_S)/(I_S[i]+S_S[i]+R_S[i])))

        M_S_end = int(np.round(100*1000*(+M_S[-1])/(N_S)))
        M_L_end = int(np.round(100*1000*(+M_L[-1])/(N_L)))
        M_end = int(np.round(100*1000*(M_L[-1]+M_S[-1])/(N_S+N_L)))
        Imune = int(100*R_S[-1]/(S_S[-1]+R_S[-1]))

#%%
```